ଓଡ଼ିଶା ରାଜ୍ୟ ମୁକ୍ତ ବିଶ୍ୱବିଦ୍ୟାଳୟ, ସମ୍ବଲପୁର, ଓଡ଼ିଶା
Odisha State Open University, Sambalpur, Odisha
Established by an Act of Government of Odisha.

ASSIGNMENT

Enrollment Number	**MAHD2420173**
Name	**JAGABANDHU PRADHAN**
Admission Session	**February 2025**
Programme	**Master of Arts (Hindi) (MAHD)**
Assignment Number	02
Course Code	PUMC -207
Course Title	CYBER SECURITY
Study Centre	Narasingh Choudhury (Auto) College, Jajpur.
Date of Submission	11/03/2026

ओड़ीशा राज्य मुक्त विश्वविद्यालय, सम्बलपुर

हिंदी विभाग

स्नातकोत्तर हिंदी

प्रश्न - पत्र

सेमेस्टर - 02

पाठ्यक्रम नाम - Cyber Security.

सत्रीय कार्य - 02

पाठ्यक्रम कोड - PUMC-207

पाठ्यक्रम क्षेत्र - इकाई 01 से 08 तक

No-1 Employees of a financial institution fall victim to a targeted spear-phishing attack leading to leakage of customer credentials and financial loss. Apply preventive and responsive cyber security measures ~~to~~ explain how such social engineering attacks can be detected, reported, and mitigated.

Introduction

In today's digital era, finacial institutions rely heavily on computer networks and online communication systems to perform daily operations. However, this increased dependence to various cyber threats. One of the most common and dangerous threats is spear-phishing, a type of social engineering attack. In a spear-phishing attack, cybercriminals send targeted and personalized emails or messages to employees to trick them into revealing sensitive information such as login credentials, banking data, or confidential documents.

When employees of a financial institution fall victim to such attacks, it can lead to serious consequences, including leakage of customer credentials, financial fraud, and reputational to implement preventive and responsive cybersecurity measures to detect report, and mitigate such attacks.

Understanding Spear-Phishing Attacks

Spear-phishing is a highly targeted form of phishing where attackers research their victims and craft convincing messages that appear to come from trusted sources. These messages may look like they are sent by managers, colleagues, banks, or official organizations.

Common characteristics of spear phishing attacks include:

(i) personalized emails addressing employees by name.

(ii) Fake login pages that mimic legitimate websites.

(iii) Urgent requests for confidential information.

(iv) Attachments containing malware.

(V) Links that redirect users to malicious websites.

In financial institutions, attackers often aim to obtain banking credentials, transaction authorization codes, or access to internal system.

Preventive Cyber Security Measures

To prevent spear-phishing attacks, Financial institutions should adopt strong cyber security practices and employee awareness programs.

1. Employee Awareness and Training

Employees are the first line of defense against phishing attacks. Organizations should conduct regular cyber security training to educate employees about phishing techniques suspicious emails, and safe internet practices. Training should include:

- Identifying fake emails and suspicious links.
- Verifying requests for sensitive information.
- Reporting suspicious activities immediately.

2. Email Security Systems

Financial institutions should implement.

advanced email filtering systems to detect and block malicious emails. Email security solutions can :

* Identify phishing attempts.
* Scan attachments for malware.
* Detect spoofed email addresses.
* Block suspicious domains.

These tools reduce the chances of phishing emails reaching employees.

3. Multi-Factor Authentication (MFA)

Multi-factor authentication adds an additional layer of security to user accounts. Even if attackers obtain login credentials, they cannot access the system without a second authentication factor such as :

- One time password (OTP)
- Biometric verification
- Security tokens

MFA significantly reduces the risk of unauthorized access.

4. Regular Security Awareness Campaigns

Organizations should conduct continuous awareness campaigns through workshops

emails, posters, and simpaigns help employees stay alert and improve their ability to detect social engineering attacks.

5. Secure Network Infrastructure

Finacial institutions should used secure firewalls, intrusion detection systems (IDS), and intrusion prevention systems (IPS) to monitor network traffic and detect suspicious activities.

Detection of Spear-Phishing Attacks

Early detection of phishing attacks is critical to prevent damage. Organizations can detect such attacks using the following methods:

1. Security Monitoring Tools

Security Information and Event Management (SIEM) systems can analyze network logs and detect unsual activities such as multiple login attempts or suspicious email traffic.

2. Threat Intelligence Systems

Threat intelligence platforms collect information about known phishing domains and malicious IP addresses. These systems help organizations block known cyber threats

before they cause harm.

3. Email Analysis

Security teams should analyze suspicious emails by checking:

- Sender's email address
- Email headers
- Embedded links
- Attachments

This helps determine whether the message is legitimate or malicious.

Reporting Mechanisms

Effective reporting procedures are essential for responding quickly to phishing attacks.

1. Internal Reporting System

Employees should be encouraged to report suspicious emails immediately to the IT security team. Many organizations provide a dedicated "Report phishing" button in email systems.

2. Incident Response Team

Organizations should have a didicated Incident Response Team (IRT) responsible for investigating and handling cybersecurity

incidents. The team should analyze the attack, identify affected systems, and take corrective actions.

3. Documentation and Analysis

All phishing incidents should be documented and analyzed to understand how the attack occurred. This information helps improve security policies and prevent similar incidents in the future.

Mitigation Strategies

If a spear-phishing attack successfully compromises employee accounts, organizations must take immediate action to minimize damage.

1. Account Lockdown

The affected accounts should be immediately disabled or locked to prevent further unauthorized access.

2. Password Reset

All compromised credentials should be reset immediately. Employees should be required to create strong passwords.

3. Malware Removal

If malicious attachments were opened, infected systems must be scanned and cleaned using antivirus and endpoint protection tools.

4. Network Isolation

Affected systems should be temporarily isolated from the network to prevent attackers from spreading malware or accessing additional systems.

5. Customer Notification

If customer data has been compromised the organization must inform affected customers and take steps to secure their accounts.

Importance of Strong Cyber Security Policies

Financial institution must develop comprehensive cybersecurity policies that clearly define employee responsibilities, security standards, and incident response procedures. ~~These~~ These policies should include:

- Password management guidelines
- Email usage policies

- Data protection rules.
- Incident Reporting Procedures

Regular audits and security assessments Should also be conducted to ensure compliance with these policies.

Conclusion

Spear-phishing attacks pose a serious threat to financial institutions because they exploit human vulnerabilites rather than technical weaknesses. When employees fall victim to such attacks, sensitive customer information and financial assets may be compromised.

To protect against these threats, organization must implement a combination of preventive and responsive cybersecurity measures. These include employee training, email security systems, multi factor authentication, security monitoring tools, and effective incident response procedures.